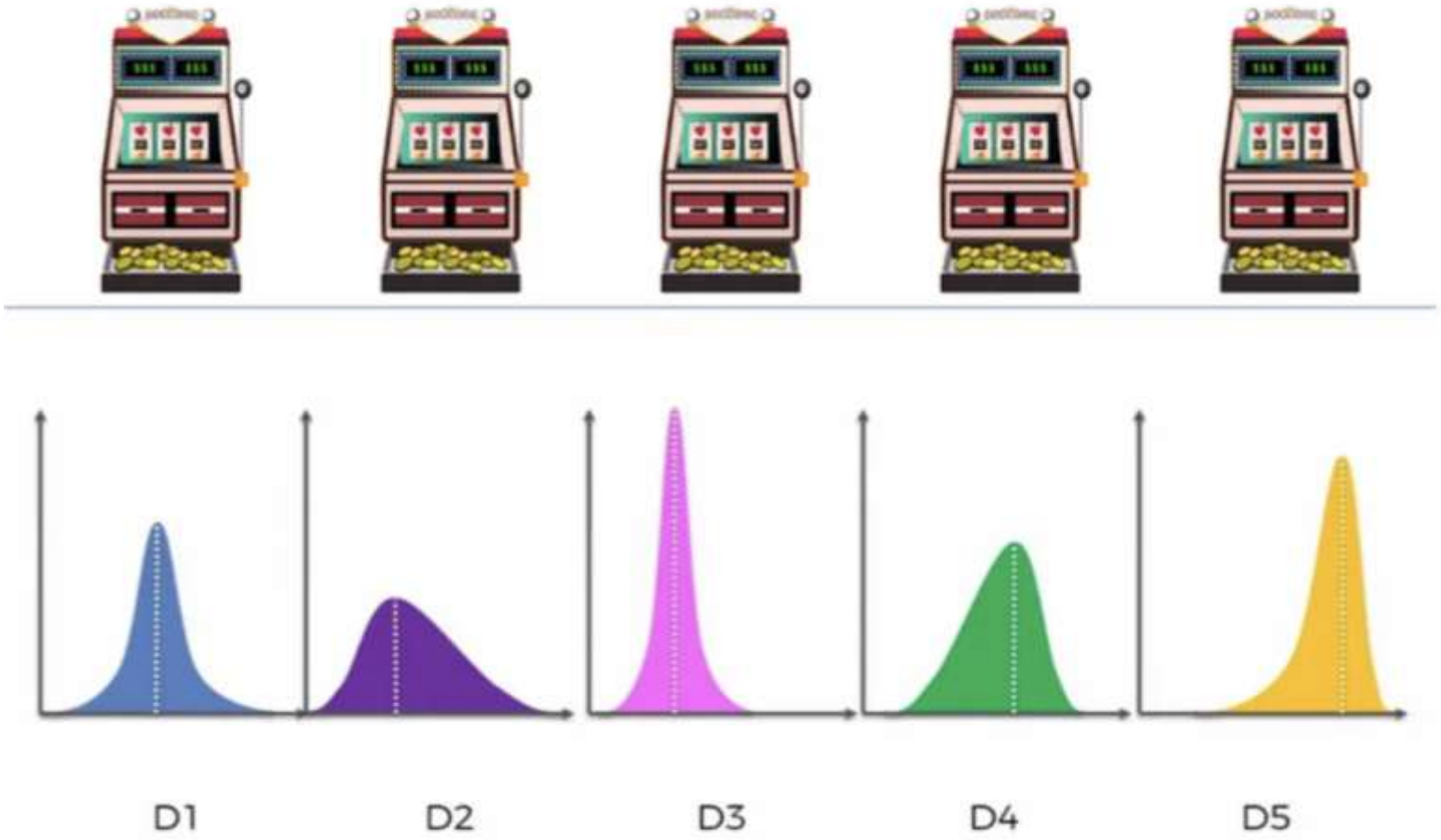
Sutton and Barto definition of RL

- “RL is learning how to map situations to actions so as to maximize a numerical reward signal. The learner is not told which actions to take, but instead must discover which actions yield the most reward by trying them. In the most challenging cases, actions effect not only the immediate reward but also the next situation, and through that, all subsequent rewards. Trial-and-error search and delayed reward are the two most important distinguishing features of RL.”
- Through trial and error, the agent gathers data about the environment. Unlike supervised and unsupervised learning, data collection is part of the RL problem.

Simplest RL problem: Multi-armed Bandit



Multi-armed bandit special case

- “RL is learning how to map situations to actions so as to maximize a numerical reward signal. The learner is not told which actions to take, but instead must discover which actions yield the most reward by trying them. ~~In the most challenging cases, actions effect not only the immediate reward but also the next situation, and through that, all subsequent rewards.~~ Trial-and-error search ~~and delayed reward~~ are the two most important distinguishing features of RL.”

The k-armed bandit problem

- k options or "actions"
- A_t = action chosen at time step t
- R_t is corresponding reward
  - Assumed random but depends on A_t
- Action value function: q(a)
- $q(a) := E[R_t \mid A_t = a]$ (we will not use textbook notation $q_*(a)$ in class)
- ***Goal:*** find $a \in \{1, \ldots, k\}$ that maximizes q(a)
  - And try to do this in as few steps as possible.

Quiz

1. Suppose we choose action a with probability $\pi(a)$. What is the $E[R_t]$ in terms of q(.) and $\pi(.)$?
2. Suppose the rewards are not random, that is, $R_t = g(A_t)$ for some fixed deterministic function g(.). What is a good algorithm for finding the optimal action.

Estimating Action-Value Function

- Let $Q_t(a)$ is an estimate of q(a) going into time step t

$$Q_t(\text{a}) := \frac{sum\ of\ rewards\ when\ a\ is\ taken\ up\ to\ t}{number\ of\ times\ a\ is\ taken\ up\ to\ t} = \sum_{i=1}^{t} R_i\, 1(A_i = \text{a})$$

- Quiz: Suppose we choose a infinitely often. What is the $\lim_{t} Q_t(a)$? By what theorem in probability theory?

Greedy Action and ϵ-Greedy Policy

- Greedy action: $A_t = \text{argmax}_a Q_t(a)$
- Quiz: Suppose we follow the following algorithm. First select each of the k actions. Then subsequently follow the greedy action.
  - After choosing the k actions, will we subsequently always choose the same action?
  - Is this policy optimal? Why or why not?

- ε-greedy policy: at each time step t:
  - choose greedy action with prob (1-ε)
  - Choose random action with probability ε

Quiz: ε-greedy policy

- With the ε-greedy policy, for the case $k=2$ and $\varepsilon = 0.1$, what is the probability that the greed action is selected?
- Generalize this result for arbitrary k and ε.

Quiz: Bandit example

Exercise 2.2: Bandit example Consider a k-armed bandit problem with $k = 4$ actions, denoted 1, 2, 3, and 4. Consider applying to this problem a bandit algorithm using ε-greedy action selection, sample-average action-value estimates, and initial estimates of $Q_1(a) = 0$, for all a. Suppose the initial sequence of actions and rewards is $A_1 = 1$, $R_1 = -1$, $A_2 = 2$, $R_2 = 1$, $A_3 = 2$, $R_3 = -2$, $A_4 = 2$, $R_4 = 2$, $A_5 = 3$, $R_5 = 0$. On some of these time steps the ε case may have occurred, causing an action to be selected at random. On which time steps did this definitely occur?

Quiz

- In supervised learning, we are given labeled data. In unsupervised learning we are given unlabeled data. What is the data in the multi-arm bandit problem?

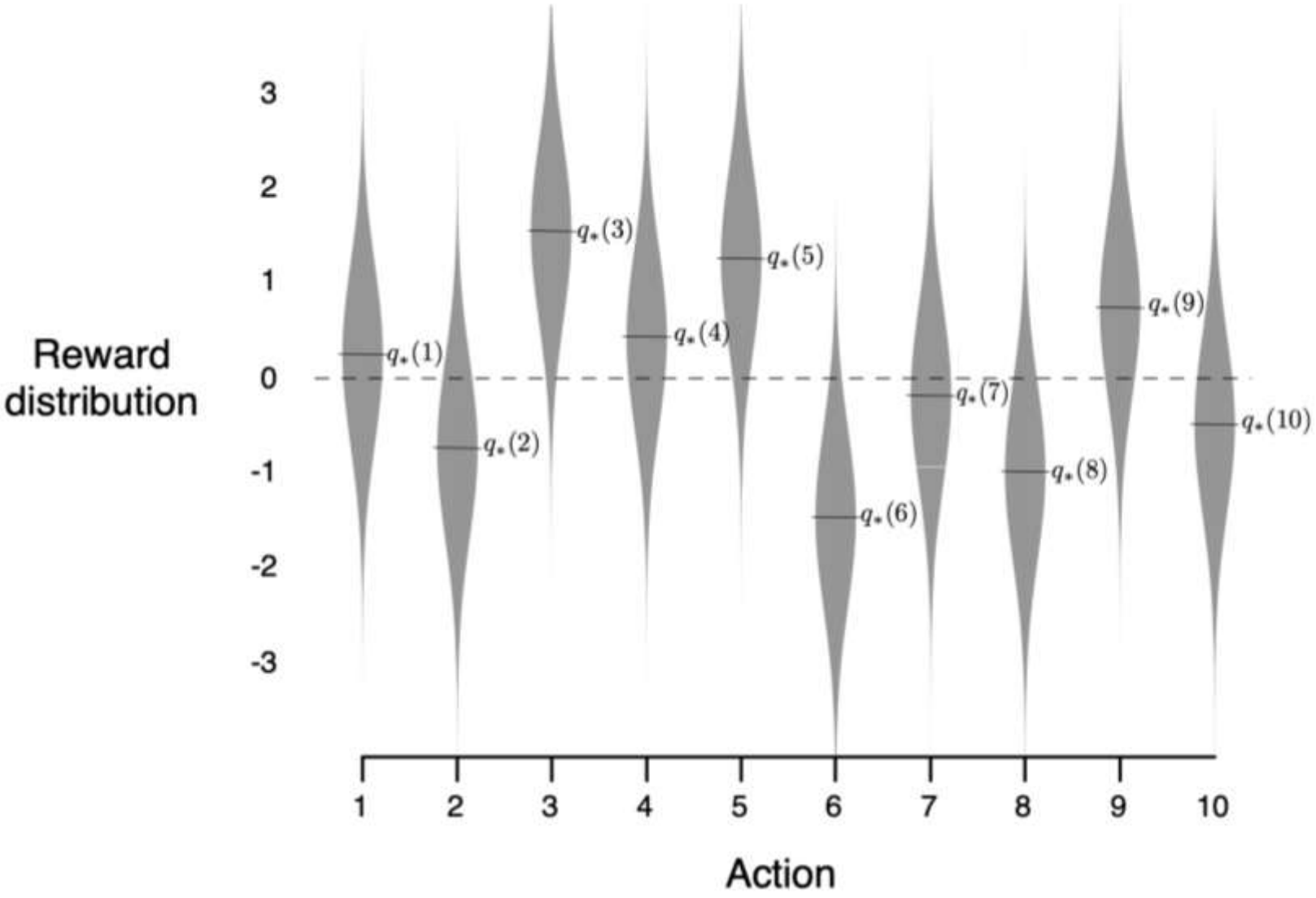
- In the ε-greedy policy, what is exploration? What is exploitation?

- Can we calculate Q_{n+1} incrementally as a function of Q_n, R_n, and n?

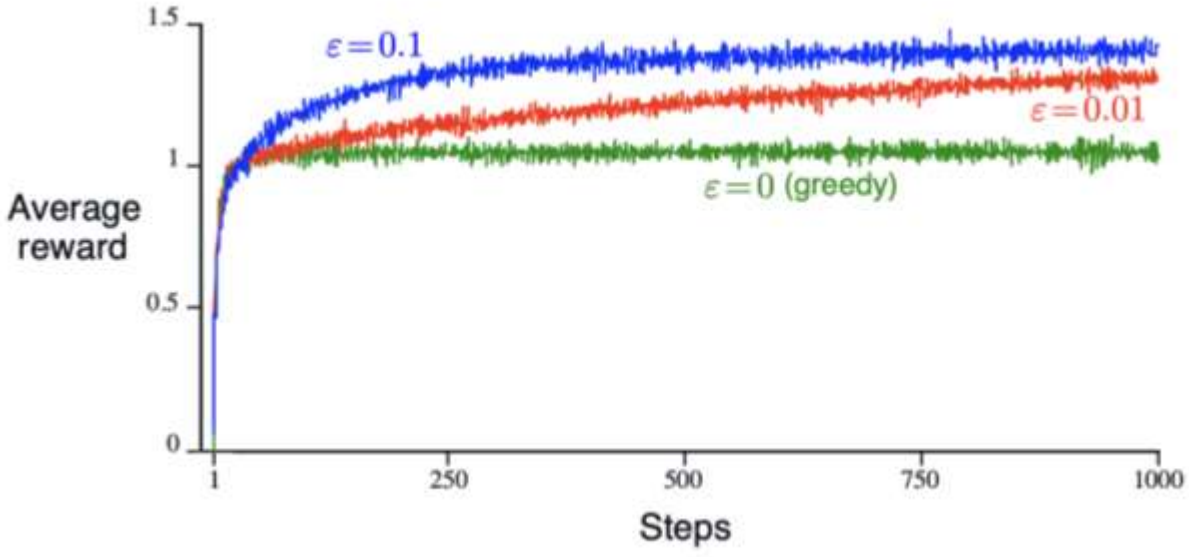
10-arm testbed

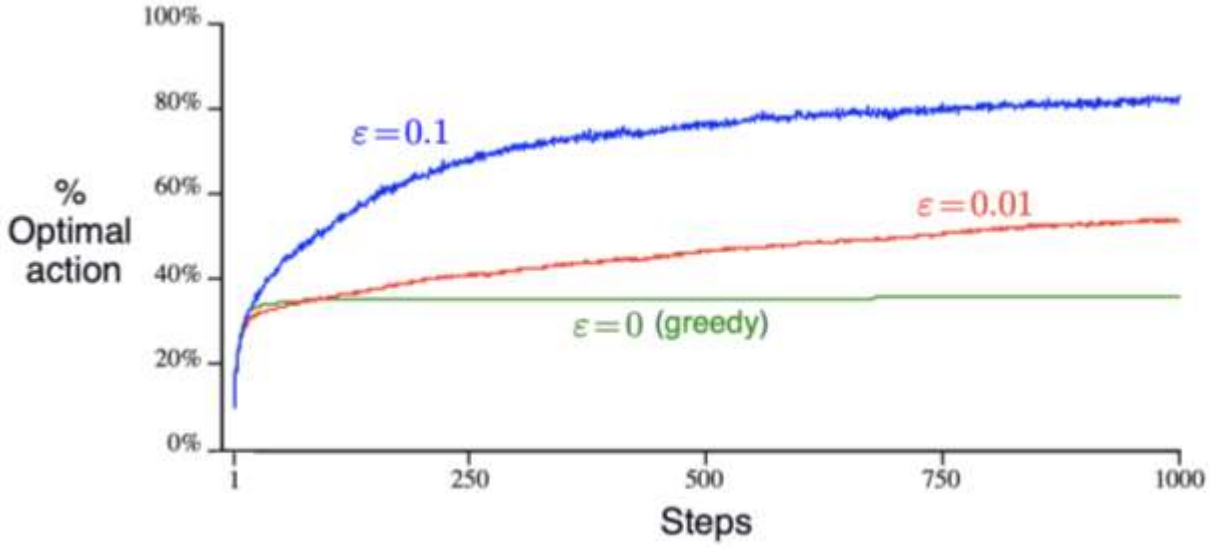
2000 randomly generated problems.

What is a randomly generated problem?



Let’s discuss these figures!





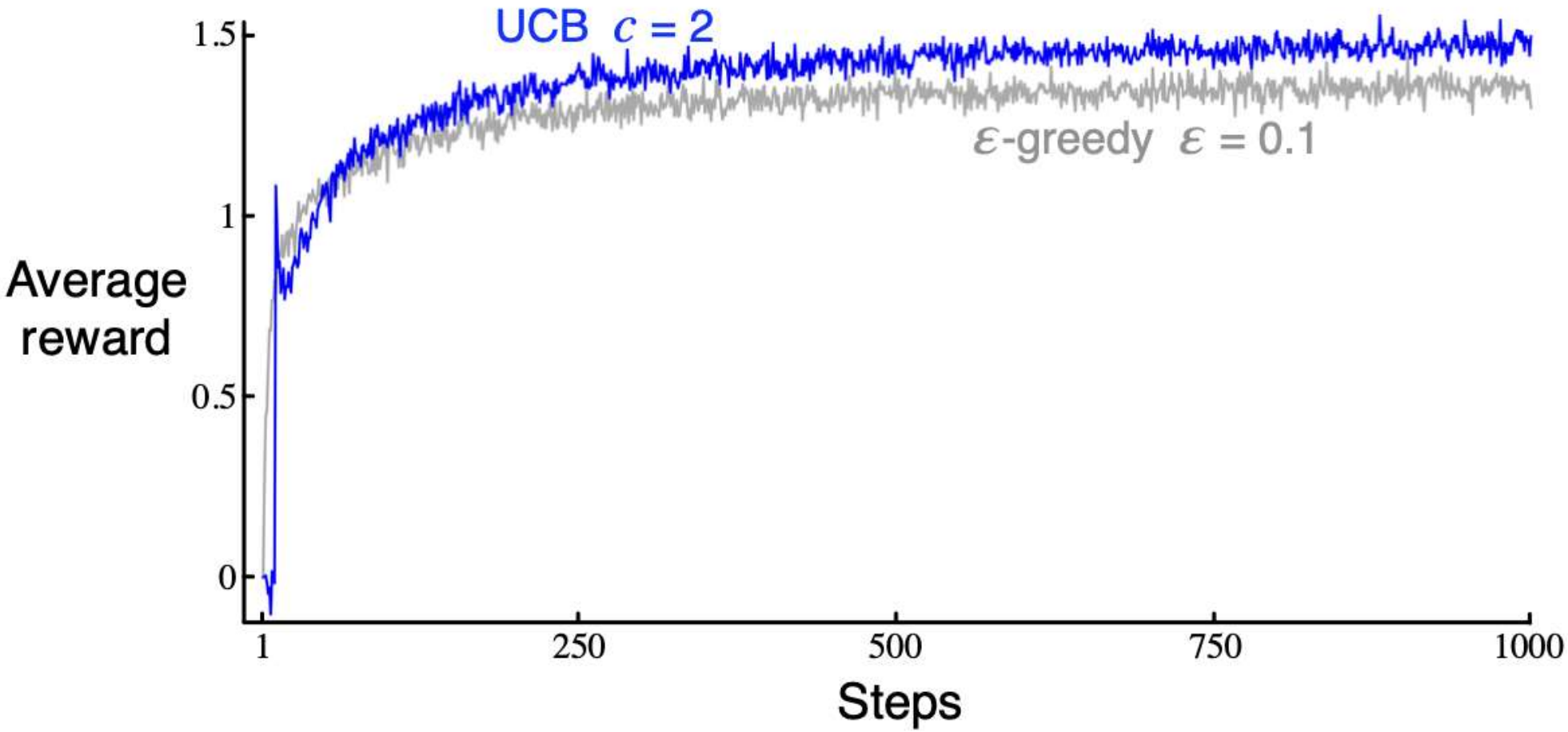
Upper Confidence Bound (UCB) Policy

$$A_t \doteq \arg\max_a \left[Q_t(a) + c\sqrt{\frac{\ln t}{N_t(a)}} \right]$$

- where $N_t(a)$ is the number of times action a selected prior to time t
- UCB algorithm is an alternative to the ε-greedy policy
- The term $c\sqrt{\frac{\ln t}{N_t(a)}}$ is a bonus term, and provides exploration
- **Question:** if action a' has been chosen less often than a'', which of the two actions will have a larger bonus?
- The bonus term is a measure of the uncertainty of our estimate: the larger the $N_t(a)$, the more certain we are about our estimate $Q_t(a)$.

Where does name “upper confidence bound” come from?

- P($q(a) \leq Q_t(a) + c\sqrt{\frac{\ln t}{N_t(a)}}$) is close to 1.
- So $Q_t(a) + c\sqrt{\frac{\ln t}{N_t(a)}}$ is essentially an upper bound on q(a)
- We would like to choose the action that maximizes q(a). But we don’t know the q(a)’s. The algorithm instead chooses the action that maximizes the upper bounds, which we do know:


UCB $c = 2$
ε-greedy $\varepsilon = 0.1$
Average reward
1.5
1
0.5
0
1
250
500
750
1000
Steps